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EDUCATION

BMS College of Engineering

B.E. Information Science & Engineering — CGPA: **9.2/10**

Bengaluru, India

2023 – 2027

EXPERIENCE

Hewlett Packard Enterprise (HPE)

Bengaluru, India

Project Intern

Feb 2026 – Present

- Built an AI-driven UEBA (User & Entity Behavior Analytics) platform for real-time threat detection, processing events from **3 log sources** (Zeek conn.log, Kafka raw-event stream, live portal auth) with **sub-50ms** end-to-end ML inference latency.
- Engineered a **Kafka** → **Elasticsearch** ingestion pipeline across **3 topics** (raw events, alerts, audit) with **2 ES indices**, partitioning indices by event type to reduce cross-index query overhead.
- Trained a **4-model weighted ensemble** (XGBoost, LightGBM, Random Forest, Gradient Boosting) on **25,000 events** with SMOTE oversampling and **46 engineered behavioral features**, achieving **98% accuracy** and **Threat-class F1 0.83** (P 0.87 / R 0.79) on a held-out 5,000-event test set.
- Automated threat response via **HashiCorp Vault AppRole** dynamic credential rotation — user credentials rotate on admin approval of BLOCK/CRITICAL alerts; PostgreSQL/Kafka credentials rotate on CRITICAL events through Vault's database secrets engine, with SMTP alerts to the security team within seconds.
- Built the **JWT-authenticated admin alert-approval API and workflow** for a real-time SOC dashboard, gating BLOCK/CRITICAL threat actions behind admin review before triggering downstream response.

Tech: Python, FastAPI, Kafka, Elasticsearch, Kibana, HashiCorp Vault, PostgreSQL, Docker, Kubernetes, MLflow, XGBoost, LightGBM

PROJECTS

Proactive ML-Driven TCP Congestion Control (Artificial ECN) | Python, LightGBM, Mininet

GitHub

- Built a real-time controller that detects congestion at F1 0.97 and flags imminent loss (~40 ms window) at F1 0.87 from live **ss** state, with ~1 ms inference inside a 20 ms poll loop; a single model generalizes across Reno/Cubic (cross-CC F1 0.94).
- Designed confidence-proportional RED/ECN queue control in Mininet that eliminated TCP Cubic retransmissions (755 → 0) and raised TCP Reno throughput ~15% in a 120s benchmark.
- Designed a leak-free multi-signal labeling strategy across 57.8K samples from 300 experiments.

Edu_RAG — Role-Based RAG Q&A System | Python, LangChain, ChromaDB, Streamlit, Ollama

GitHub

- Built a hybrid retrieval pipeline combining BM25 and dense vector search (weighted 50/50) over ChromaDB, with FlashRank cross-encoder reranking to narrow retrieved candidates to the top-ranked context per query.
- Implemented multimodal document ingestion (PDF, DOCX, PPTX) with Tesseract OCR and structured PyMuPDF parsing, plus semantic chunking (400-token child chunks) for diagram- and table-heavy course notes.
- Added role-based access (admin/student) with session-timeout security, conversational summary-buffer memory, and a heuristic answer-verification layer (0.8 coverage threshold); served locally via Ollama (Llama 3.1 8B) in a Dockerized deployment.

Real-Time AI Exam Proctoring System | Python, OpenCV, InsightFace, YOLOv8, WebSockets

GitHub

- Built face-identity verification (cosine similarity on ArcFace embeddings, 0.4 threshold), real-time phone detection (YOLOv8n), and a weighted attention scoring engine combining head pose, iris-based gaze, and face presence with EMA smoothing.
- Designed an async WebSocket server using **asyncio.to_thread** to offload ML inference, with per-event-type cooldowns and frame-skipping (face analysis every 4th frame) to sustain real-time processing on CPU.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, SQL

Backend: REST APIs, FastAPI, Node.js, Express, WebSockets

Data & ML: scikit-learn, XGBoost, LightGBM, OpenCV, Pandas, NumPy, Feature Engineering, Model Evaluation

Infrastructure: Docker, Kafka, Elasticsearch, HashiCorp Vault, Git, Github Actions(CI/CD)

Databases: PostgreSQL, MySQL, MongoDB

ACHIEVEMENTS & CERTIFICATIONS

- Winner, HyperAPI Hackathon by Hyperbots Inc.** (2026) — built an end-to-end financial document processing pipeline (PDF extraction → field structuring → GST/invoice/transaction validation → anomaly detection for duplicates and mismatches), selected from **50+ teams / 100+ participants**.